The Gravitational Refractive Index as Petz Recovery Fidelity: Temporal Asymmetry from Spacetime Geometry

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In a companion paper (Paper I), we established the temporal asymmetry parameter $\tau = 1-F$ from the Petz recovery framework and the universal bound $F \geq e^{-\Sigma/2}$. Here we apply this framework to gravitational fields. Defining the gravitational entropy production $\Sigma_{\rm grav} \equiv 2|\Phi|/c^2 = r_{\rm s}/r$, we obtain the identification $c_{\rm eff}/c = e^{\Phi/c^2} = e^{-\Sigma_{\rm grav}/2} = F_{\rm bound}$: the coordinate speed of light *is* the Petz recovery fidelity bound [NEW SYNTHESIS]. This connects Dicke's gravitational refractive index to quantum information recovery. Three independent derivations—modular flow, gravitational Landauer principle, and quantum channel analysis—converge on $\Sigma_{\rm grav} = -\ln(-g_{00})$. Saturation of the Petz bound uniquely determines the exponential metric $g_{00} = -e^{-r_{\rm s}/r}$, which has no event horizon ($\tau < 1$ everywhere). We extend the framework to gravitational decoherence (Pikovski mechanism), the Unruh effect, Bekenstein bound, black hole complementarity, the Page curve, the Hayden–Preskill protocol, and ER=EPR, classifying each result as [KNOWN], [NEW SYNTHESIS], [NEW DEFINITION], or [SPECULATIVE]. Testable strong-field predictions are tabulated.

INTRODUCTION

In Paper I [1], we established three results. First, the Petz recovery map [3] is the unique Bayesian retrodiction functor [6]. Second, the temporal asymmetry parameter $\tau \equiv 1 - F(\rho, \mathcal{R}_{\sigma,\mathcal{N}} \circ \mathcal{N}(\rho)) \in [0,1]$ quantifies the arrow of time: $\tau = 0$ for closed (unitary) evolution, $\tau > 0$ for open systems, with the universal bound [4, 5]

$$\tau \;\leq\; 1 - e^{-\Sigma/2}\,, \tag{1}$$

where Σ is the entropy production of the channel $\mathcal{N}$. Third, the equivalence chain $\tau = 0 \Leftrightarrow \Sigma = 0 \Leftrightarrow I(A;E|B) = 0 \Leftrightarrow$ QMC connects quantum erasure, error correction, and thermodynamic irreversibility.

General relativity admits a remarkable reformulation: gravity acts as a medium with position-dependent refractive index [7, 8]. The coordinate speed of light in a weak gravitational field is $c_{\rm eff}/c = e^{\Phi/c^2}$, where $\Phi < 0$ is the Newtonian potential. In this paper we show that this expression is not merely a kinematic accident but a manifestation of the Petz recovery bound:

$$\frac{c_{\rm eff}}{c} \;=\; e^{\Phi/c^2} \;=\; e^{-\Sigma_{\rm grav}/2} \;=\; F_{\rm bound}\,. \tag{2}$$

Light slowing down in a gravitational field *is* information loss; the Petz recovery fidelity *is* the refractive index.

THE EXPONENTIAL IDENTIFICATION

The coordinate speed of light

[KNOWN] In the post-Newtonian approximation of GR, the coordinate speed of light in a static gravitational field is [7, 8]

$$\frac{c_{\rm eff}}{c} = e^{\Phi/c^2}\,, \tag{3}$$

where $\Phi < 0$ is the Newtonian gravitational potential. This result, first anticipated by Einstein in 1911 [7] and placed on rigorous footing by Dicke [8], is exact for the exponential metric and agrees with the Schwarzschild metric through $O(r_{\rm s}/r)^2$.

Gravitational entropy production

[NEW DEFINITION] We define the dimensionless gravitational entropy production as

$$\boxed{\Sigma_{\rm grav} \;\equiv\; \frac{2|\Phi|}{c^2} \;=\; \frac{r_{\rm s}}{r}\,,} \tag{4}$$

where $r_{\rm s} = 2GM/c^2$ is the Schwarzschild radius. This is an *informational* entropy production—the cost of retrodiction from infinity to radius r—not a thermodynamic entropy in the Clausius sense. The Clausius entropy production for a photon in Tolman thermal equilibrium vanishes identically [20], because $T(\Phi)\sqrt{-g_{00}} = T_\infty$. By contrast, $\Sigma_{\rm grav}$ quantifies the informational cost of climbing the gravitational potential, which persists even in equilibrium.

Three independent arguments converge on Eq. (4):

(a) Modular flow [KNOWN].—In the entanglement first-law regime of AQFT on Schwarzschild, the fractional quantum relative entropy (QRE) loss between radius r and infinity is $(D_{\rm in} - D_{\rm out})/D_{\rm in} = r_{\rm s}/r$, exactly, via the Tolman blueshift and energy redshift [24].

(b) Gravitational Landauer principle [KNOWN].—The Tolman-modified cost of erasing one bit at radius r exceeds that at infinity by a factor $1/\sqrt{-g_{00}}$. The dimensionless excess erasure cost yields $\Sigma_{\rm grav} = -\ln(-g_{00})$ [25], which equals $r_{\rm s}/r$ for the exponential metric and $-\ln(1 - r_{\rm s}/r) \approx r_{\rm s}/r + O((r_{\rm s}/r)^2)$ for Schwarzschild.

(c) Quantum channel model [KNOWN].—Modeling gravitational redshift as a pure-loss bosonic channel with intensity transmissivity $\eta = -g_{00}$, the QRE decrease is $\Sigma = -\ln\eta = r_s/r$ [26].

All three routes share the common mathematical core $\Sigma_{\rm grav} = -\ln(-g_{00})$.

The channel problem

[NEW SYNTHESIS] None of the three routes above constructs a canonical gravitational CPTP map $\mathcal{N}_{\rm grav}$ from first principles—i.e., a map with explicit Kraus operators $\{K_i\}$ such that $\Sigma = D(\rho\|\sigma) - D(\mathcal{N}_{\rm grav}(\rho)\|\mathcal{N}_{\rm grav}(\sigma)) = r_s/r$. This remains an open problem [**?**]. We bypass it through a combined strategy: (i) any consistent coupling of classical gravity to quantum matter is fundamentally irreversible ($\Sigma > 0$), guaranteeing the *existence* of an irreversible CPTP map [**?**]; (ii) the modular flow route (a) provides the coefficient r_s/r without requiring an explicit channel [24]; (iii) the pure-loss bosonic channel (c) serves as a concrete *existence proof* that a CPTP map with $\Sigma = -\ln\eta = r_s/r$ exists when $\eta = -g_{00}$; (iv) axiomatic uniqueness: any CPTP map satisfying additivity under channel composition and monotonicity under coarse-graining yields $\Sigma = -c\,\ln(-g_{00})$ with $c > 0$; routes (a)–(c) fix $c = 1$. The recent modular channel construction of Trejo-Calderón [**?**]—a CPTP map built from Gibbs-weighted singular values of the modular operator—is the most promising candidate for a first-principles $\mathcal{N}_{\rm grav}$, though the explicit computation of ΔD in Schwarzschild remains to be carried out.

The triple identification

[NEW SYNTHESIS] Substituting Eq. (4) into the Petz fidelity bound $F \geq e^{-\Sigma/2}$ from Paper I:

$$F_{\rm bound} = e^{-\Sigma_{\rm grav}/2} = e^{\Phi/c^2}\,. \tag{5}$$

Comparing with Eq. (3), we obtain the central result:

$$\boxed{\frac{c_{\rm eff}}{c} \;=\; e^{-\Sigma_{\rm grav}/2} \;=\; F_{\rm bound}\,.} \tag{6}$$

The effective speed of light *is* the Petz recovery fidelity bound. Equivalently, the gravitational refractive index is the inverse fidelity:

$$n_{\rm grav}(r) = \frac{c}{c_{\rm eff}} = e^{+\Sigma_{\rm grav}/2} = \frac{1}{F_{\rm bound}}\,. \tag{7}$$

Table I provides numerical verification across four orders of magnitude in gravitational compactness.

Caveat.—The identification (6) is exact for the exponential metric $g_{00} = -e^{-r_s/r}$ and agrees with Schwarzschild through $O(r_s/r)$. At $O((r_s/r)^2)$, the two metrics differ: the Schwarzschild expansion gives $g_{00}^{\rm Schw} = -1 + r_s/r - r_s^2/(2r^2) + \cdots$, identical to the exponential expansion through this order. The first discrepancy appears at $O((r_s/r)^3)$. In the solar system ($r_s/r \sim 10^{-6}$), this cubic correction is of order 10^{-18}—far below current measurement capability [13].

TABLE I. Gravitational entropy production $\Sigma_{\rm grav} = r_s/r$ and Petz recovery fidelity bound $F_{\rm bound} = e^{-\Sigma_{\rm grav}/2}$ for representative astrophysical objects. The last row indicates the limit of the weak-field approximation.

Object	r_s/r	$F_{\rm bound}$	$\tau_{\max}$
Earth surface	1.4×10^{-9}	0.99999999930	7.0×10^{-10}
Sun surface	4.2×10^{-6}	0.9999979	2.1×10^{-6}
Neutron star	0.4	0.819	0.181
Schwarzschild $r=r_s$	1	0.607	0.393

GRAVITATIONAL DECOHERENCE AS PETZ RECOVERY

The Pikovski mechanism

[KNOWN] Pikovski et al. [19] showed that gravitational time dilation induces universal dephasing of massive objects in spatial superposition. The Hamiltonian $H = p^2/(2m) + H_{\rm int}(1 + \Phi(\hat{x})/c^2)$ produces a position-dependent phase on the internal state, yielding a CPTP dephasing channel $\mathcal{N}_{\rm grav}$ on the spatial degree of freedom [2].

[NEW SYNTHESIS] For N internal modes in a thermal state, the temporal asymmetry parameter is [2]

$$\tau_{\rm grav}(t) = 1 - \exp\big(-N\Gamma t^2\big)\,, \tag{8}$$

where $\Gamma = (\Delta E)^2(\Delta\Phi)^2/(2\hbar^2c^4)$ and ΔE is the internal energy variance. The connection to Sec. is direct: the $\Sigma_{\rm grav} = r_s/r$ from the gravitational potential provides the entropy production that bounds $\tau_{\rm grav}$ via Eq. (1). The dephasing rate is controlled by $\Delta\Phi/c^2 \sim \Sigma_{\rm grav}/2$, establishing that the gravitational entropy production governs both the static (refractive index) and dynamic (decoherence) aspects of gravitational information loss.

Unruh effect

[NEW SYNTHESIS] The Unruh effect [22] states that a uniformly accelerating observer with acceleration a perceives a thermal bath at temperature $T_U = \hbar a/(2\pi c k_{\rm B})$. This thermal environment induces entropy production on any quantum system carried by the observer. For a system traversing a proper distance Δx at acceleration a,

the dimensionless entropy production is [21]

$$\Sigma_{\text{Unruh}} = \frac{2\pi a\,\Delta x}{\hbar c^2}\,, \tag{9}$$

giving $\tau > 0$ for any accelerating observer. By the equivalence principle, this connects to gravitational entropy production: a static observer at radius r experiences proper acceleration $a = GM/(r^2\sqrt{1 - r_{\text{s}}/r})$, and the accumulated entropy production over radial displacement Δr reproduces Σ_{grav} in the appropriate limit.

Bekenstein bound in τ language

[NEW SYNTHESIS] The Bekenstein bound [23] states that the entropy of a system of energy E confined to a sphere of radius R satisfies $S \leq 2\pi ER/(\hbar c)$. In the τ framework, this becomes

$$\tau \;\leq\; 1 - \exp\!\left(-\frac{\pi ER}{\hbar c}\right), \tag{10}$$

where we have used $\Sigma_{\text{Bek}} = 2\pi ER/(\hbar c)$ and the Petz bound $\tau \leq 1 - e^{-\Sigma/2}$. This provides a universal upper limit on the temporal asymmetry of any spatially bounded quantum system.

BLACK HOLE PHYSICS THROUGH τ

Complementarity

[NEW SYNTHESIS] Black hole complementarity posits that infalling and external observers describe different but consistent physics [27]. In the τ framework, this becomes quantitative: the infalling observer follows a geodesic and experiences $\tau_{\text{infalling}} \approx 0$ (locally, spacetime is flat by the equivalence principle), while the external observer describes the same process with $\tau_{\text{external}} \to 1$ (information appears to be lost at the horizon). Complementarity is thus the statement that different quantum channels—corresponding to different observer foliations—assign different τ values to the same physical process. The Petz recovery map is observer-dependent: the recovery available to the infalling observer (who has direct access to the interior) differs from that available to the external observer (who can only perform operations on Hawking radiation).

Page curve as $\tau(t)$

[NEW SYNTHESIS] The Page curve describes the entanglement entropy $S_{\text{Page}}(t)$ of Hawking radiation as a function of time [28]. Before the Page time, entropy increases linearly; after the Page time, it decreases, returning to zero when evaporation is complete. In the τ framework, the Page curve translates to a time-dependent bound on recovery:

$$\tau(t) \;\leq\; 1 - \exp\!\big(-S_{\text{Page}}(t)/2\big)\,. \tag{11}$$

Before the Page time, S_{Page} increases and τ can grow—information becomes progressively harder to recover. After the Page time, S_{Page} decreases and the bound on τ tightens—information recovery improves. The Page curve is thus the trajectory of the system through τ space, and the Page time marks the maximum of temporal asymmetry.

Recent developments [29, 30] have provided explicit calculations of the Page curve using the quantum extremal surface (QES) formula, confirming that $S_{\text{Page}}(t)$ follows the Page curve and that unitarity is preserved. In our framework, the QES formula for generalized entropy S_{gen} provides the entropy production $\Sigma = \Delta S_{\text{gen}}$ that enters the Petz bound.

Hayden–Preskill = Petz recovery

[KNOWN] Cotler et al. [32] proved that the Hayden–Preskill decoder [31]—the protocol for recovering information thrown into an old black hole—*is* the Petz recovery map applied to the black hole channel. This is not an analogy: the HP decoder is mathematically identical to the rotated Petz map $\mathcal{R}_{\sigma,\mathcal{N}}$ [5]. The fidelity of information recovery satisfies the same bound $F \geq e^{-\Sigma/2}$, where Σ is the entropy production of the black hole evaporation channel.

Entanglement wedge reconstruction

[KNOWN] In AdS/CFT, bulk quantum fields are reconstructed from boundary data using the Petz recovery map [32–34]. The fidelity of entanglement wedge reconstruction satisfies [29, 35]

$$F^2 \;\geq\; \exp(-\Delta S_{\text{gen}})\,, \tag{12}$$

where ΔS_{gen} is the change in generalized entropy. This is an instance of the same universal recovery inequality [5] that underlies our gravitational identification (6).

ER=EPR

[SPECULATIVE] The ER=EPR conjecture [36] identifies Einstein–Rosen bridges (wormholes) with Einstein–Podolsky–Rosen correlations (entanglement). In the τ framework, a wormhole connecting two entangled black holes provides a channel with $\tau = 0$ between the two sides: information can be perfectly recovered through

TABLE II. Classification of results connecting the τ framework to black hole physics.

Connection	Status
HP decoder = Petz map [32]	KNOWN
Entanglement wedge via Petz [34]	KNOWN
$F^2 \geq e^{-\Delta S_{\rm gen}}$ [29]	KNOWN
Observer-dependent τ (complementarity)	NEW SYNTHESIS
Page curve as $\tau(t)$	NEW SYNTHESIS
$c_{\rm eff}/c = F_{\rm bound}$ (refractive index)	NEW SYNTHESIS
$\Sigma_{\rm grav} = r_{\rm s}/r$ (gravitational entropy production)	NEW DEFINITION
ER=EPR as $\tau = 0$ channel	SPECULATIVE

the bridge, corresponding to the maximal entanglement of the two black holes. The non-traversability of the classical ER bridge corresponds to the fact that the $\tau = 0$ channel requires operations on both sides simultaneously (entanglement does not enable superluminal signaling).

Classification

Table II classifies the connections developed in this section.

THE EXPONENTIAL METRIC

[NEW SYNTHESIS] If the Petz bound is *saturated*—i.e., the recovery fidelity equals rather than exceeds $e^{-\Sigma_{\rm grav}/2}$—then the metric component is determined: $\sqrt{-g_{00}} = e^{-\Sigma_{\rm grav}/2} = e^{-r_{\rm s}/(2r)}$. This uniquely determines the exponential metric [9, 10]:

$$ds^2 = -e^{-r_{\rm s}/r}\, c^2 dt^2 + e^{+r_{\rm s}/r}\left(dr^2 + r^2 d\Omega^2\right), \qquad (13)$$

in isotropic coordinates.

No event horizon.—Since $e^{-r_{\rm s}/r} > 0$ for all $r > 0$, the metric component g_{00} never vanishes. Equivalently, $\tau < 1$ everywhere: complete information loss never occurs. This is the information-theoretic statement that event horizons are forbidden when $\Sigma_{\rm grav} < \infty$—a model-independent consequence of the Petz bound [5] that does not depend on the saturation hypothesis.

Exact solution.—The exponential metric is not a vacuum solution of the Einstein equations ($R_{\mu\nu} \neq 0$). It is an exact solution of Einstein gravity coupled to a phantom scalar field $\phi = M/r$ [14], and also arises in logarithmic $f(R)$ gravity [15] and scalar-Einstein-Gauss-Bonnet theory [16]. Makukov and Mychelkin [11] showed that three independent scalar-field families within GR all reduce to the exponential metric when the scalar charge equals the gravitational mass. Boonserm et al. [12] classified the geometry as a traversable wormhole.

The spatial metric is assumed, not derived.—The information-theoretic framework constrains only the temporal component g_{00} through $\sqrt{-g_{00}} = e^{-\Sigma_{\rm grav}/2}$. The spatial component $g_{rr} = e^{+r_{\rm s}/r}$ in Eq. (13) follows from the isometry condition $g_{00} \cdot g_{rr} = -1$ (isotropic coordinates), which is equivalent to Dicke's scalar refractive index [8]: $n^2 = 1/(-g_{00})$ with $g_{rr} = n^2$. This is a physically motivated assumption—not a derivation from quantum information theory. The fundamental reason is that quantum channels model *temporal* processes (evolution, decoherence, loss), and the Petz recovery framework naturally constrains the temporal metric component; no analogous information-degradation process exists for the spatial direction. Consequently, we classify our predictions as *robust* (depending only on g_{00}: no-horizon theorem, refractive index, gravitational decoherence rate) or *conditional* (depending on the full metric: shadow size, ISCO radius, QNM frequencies, echo delay). Note that the Jacobson [38] and Dorau–Much [24] programs *do* derive all metric components from information-theoretic premises, but their vacuum solution is Schwarzschild, not exponential. This tension between derivability and the specific metric form remains an open question.

Comparison with Schwarzschild.—The exponential and Schwarzschild metrics share identical PPN parameters $\gamma = \beta = 1$ and agree through $O(r_{\rm s}/r)^2$:

$$g_{00}^{\rm Schw} = -1 + \frac{r_{\rm s}}{r} - \frac{r_{\rm s}^2}{2r^2} + \cdots, \qquad (14)$$

$$g_{00}^{\rm exp} = -1 + \frac{r_{\rm s}}{r} - \frac{r_{\rm s}^2}{2r^2} + \cdots. \qquad (15)$$

The first difference appears at $O(r_{\rm s}/r)^3$: $-1/6$ (exponential) versus $-1/4$ (Schwarzschild in isotropic coordinates). All solar-system tests are passed automatically, including the Cassini bound $|\gamma - 1| < 2.3 \times 10^{-5}$ [13].

Observational constraints.—In the strong-field regime, the exponential metric predicts a photon sphere at $r = r_{\rm s}$ (isotropic coordinates), a shadow 4.6% larger than Schwarzschild [12], QNM frequencies shifted by $\sim 4.4\%$, and gravitational-wave echoes at $\Delta t \approx 4.2\, r_{\rm s}/c$ [41]. The EHT measurement of 42 ± 3 μas for M87* [39] is consistent with both metrics. NICER X-ray observations [40] and next-generation gravitational-wave detectors can probe the $\sim$ 5–19% strong-field deviations.

Honest assessment.—The strong-field regime is speculative. The saturation hypothesis ($F = e^{-\Sigma_{\rm grav}/2}$ exactly) is unproven; if nature selects $F > e^{-\Sigma_{\rm grav}/2}$, the actual metric lies between the exponential and Schwarzschild predictions. Current data cannot distinguish the two metrics: the most distinctive prediction—gravitational-wave echoes at $\Delta t \approx 4.2\, r_{\rm s}/c$ (~ 2.5 ms for $60\, M_\odot$)—has not been searched for, as existing echo analyses use ~ 200 ms templates optimized for Planck-wall models [41]. Dedicated short-delay searches using existing LIGO–Virgo O3/O4 data could test this prediction.

The identification (6) is robust in weak fields and suggestive in strong fields. All conditional predictions (shadow, ISCO, QNM, echoes) depend on the isometry assumption $g_{rr} = e^{+r_s/r}$; the robust predictions (no-horizon theorem, $n_{\rm grav} = 1/F_{\rm bound}$, decoherence bound) depend only on g_{00}.

WHAT τ CAN AND CANNOT EXPLAIN

Can:

- Unify quantum error correction, thermodynamic irreversibility, black hole information, and gravitational decoherence under a single parameter τ.
- Provide universal bounds: any gravitational channel satisfies $F \geq e^{-\Sigma_{\rm grav}/2}$, constraining recovery independently of the specific protocol.
- Reproduce the Bekenstein bound, the Unruh entropy production, and the Page curve bound in τ language.
- Identify the Hayden–Preskill decoder as the Petz map (established result [32]).
- Derive the no-horizon theorem: $\tau < 1$ for any finite-mass object at finite distance (model-independent [5]).

Cannot:

- Explain dark matter: at galactic scales, $\Sigma_{\rm grav} \sim 10^{-6}$, far too small to produce observable effects beyond Newtonian gravity.
- Explain dark energy or the cosmological constant: the framework is silent on cosmological dynamics.
- Derive g_{ab} from τ: the relation is one-directional ($g \to \Sigma \to \tau$); τ constrains only g_{00}, not the full metric tensor. The spatial component g_{rr} requires an independent assumption (Sec.); programs that derive all components (Jacobson, Dorau–Much) yield Schwarzschild, not the exponential metric.
- Construct a canonical gravitational CPTP map: the bypass strategy (Sec.) establishes $\Sigma_{\rm grav} = r_s/r$ without an explicit channel, but the Kraus-operator representation of $\mathcal{N}_{\rm grav}$ remains open [? ?].
- Resolve the firewall paradox [37]: while $\tau < 1$ removes the event horizon premise, the framework does not address the interior structure of compact objects.
- Apply to general dynamical spacetimes without modification: the static formulation requires a timelike Killing vector (see Discussion for the Kodama extension).

DISCUSSION AND OUTLOOK

The identification $c_{\rm eff}/c = F_{\rm bound}$ (Eq. (6)) provides a concrete bridge between quantum information theory and gravitational physics. Its logical structure is complementary to the Jacobson program [38]: while Jacobson derives *field equations* from thermodynamic entropy, we derive the *metric* directly from information recoverability. Both approaches share the premise that geometry is information-theoretic in origin; they differ in which information-theoretic quantity is taken as fundamental and consequently arrive at different strong-field predictions.

The Dorau–Much result [24]—deriving the semiclassical Einstein equations from quantum relative entropy—suggests that the two programs may converge: our $\Sigma_{\rm grav} = -\ln(-g_{00})$ and their QRE-based field equations share the same informational foundation.

Several open questions merit investigation.

(i) *Dynamic spacetimes.*—The most pressing limitation is the restriction to static metrics. The Kodama vector [?]—the unique divergence-free timelike vector in spherically symmetric spacetimes—provides a natural extension: $F^K_{\rm bound}(A \to B) = |K|_A/|K|_B$. In the static limit, K^a reduces to the Killing vector and $F^K_{\rm bound} = \sqrt{-g_{00}}$, recovering all results of this paper. In Vaidya collapse (mass accretion $M(v)$), $\tau_K(r,v) = 1 - \sqrt{1-2M(v)/r}$, tracking the *apparent* horizon—a quasi-local, causal surface—rather than the event horizon. In FRW cosmology, $\tau_K(R) = 1-\sqrt{1-H^2R^2}$, reaching $\tau = 1$ at the Hubble radius. Recent work extends the Kodama approach to axisymmetric spacetimes [?]. The deeper theoretical foundation lies in modular flow [? ?]: the Tomita–Takesaki modular automorphism defines τ for any spacetime and any state, without requiring symmetry. Crossed-product algebras make gravitational entropy well-defined and observer-dependent [?], consistent with the coordinate dependence of $\Sigma_{\rm grav}$. Cirafici [?] has established fluctuation theorems in gravitational algebras, confirming that the Crooks–Petz connection of Paper I extends to de Sitter spacetime.

(ii) *τ at the Planck scale*: when $\Sigma_{\rm grav} \sim O(1)$, quantum gravity effects become important. Loop quantum gravity corrections to the exponential metric have been computed [43], but their information-theoretic interpretation remains unexplored.

(iii) *The canonical channel*: the modular channel construction of Trejo-Calderón [?] builds a CPTP map from the singular values of the modular operator. Computing its QRE loss in Schwarzschild would either confirm $\Sigma = r_s/r$ from first principles or reveal a correction. Kasprzak and Tjoa [?] provide the most rigorous framework for quantum information transmission through curved-spacetime fields; combining their channel with the modular construction is a concrete next step.

(iv) *Rotation*: the Kerr analog of the exponential metric [17, 18] has neither horizons nor ergospheres; its information-theoretic derivation requires extending the framework beyond spherical symmetry.

(v) *Non-Markovian gravity*: if the gravitational channel exhibits memory effects, the entropy production Σ_{grav} may not be additive, and the Petz bound requires modification [21].

In Paper 1b [2], we applied the τ framework to wavefunction collapse, showing that gravitational decoherence drives $\tau \to 1$ and objectivity ($R \to N$) at the same timescale. The present paper extends the framework to strong gravitational fields. A future paper (Paper 3) will investigate the weak-field limit at galactic scales, where quantum corrections to the gravitational channel may produce observable effects on rotation curves.

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